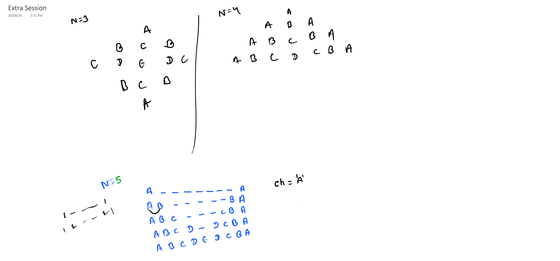


Is Armstrong Number

Take the following as input.

A number

Write a function which returns true if the number is an armstrong number and false otherwise, where Armstrong number is defined as follows.

A positive integer of n digits is called an Armstrong number of order n (order is number of digits) if.

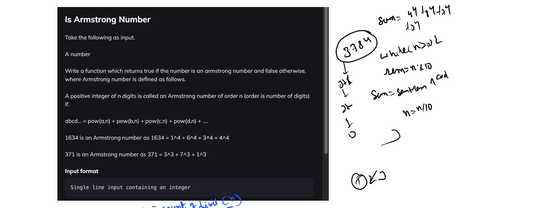
abcd... = pow(a,n) + pow(b,n) + pow(c,n) + pow(d,n) + ...

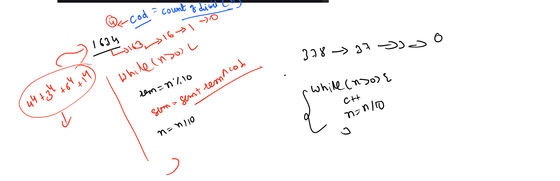
1634 is an Armstrong number as 1634 = 1^4 + 6^4 + 3^4 + 4^4

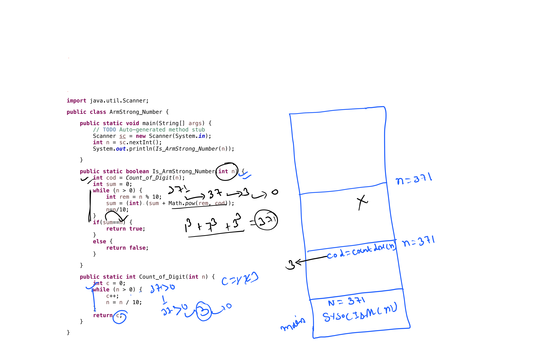
371 is an Armstrong number as 371 = 3^3 + 7^3 + 1^3

Input format

```
Single line input containing an integer
```

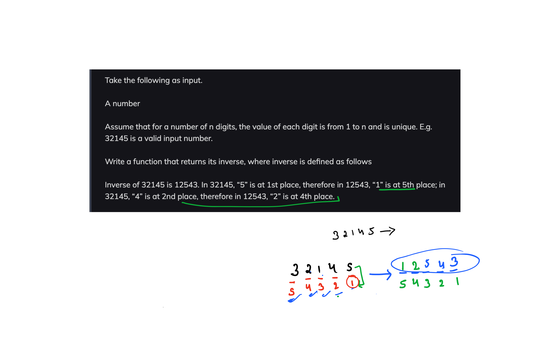






```java
import java.util.Scanner;
public class ArmStrong_Number {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        System.out.println(Is_ArmStrong_Number(n));
    }
    public static boolean Is_ArmStrong_Number(int n) {
        int cod = Count_of_Digit(n);
        int sum = 0;
        while (n > 0) {
            int rem = n % 10;
            sum = (int) (sum + Math.pow(rem, cod));
            n = n / 10;
        }
        if(sum==n) {
            return true;
        }
        else {
            return false;
        }
    }
    public static int Count_of_Digit(int n) {
        int c = 0;
        while (n > 0) {
            c++;
            n = n / 10;
        }
        return c;
    }
}
```



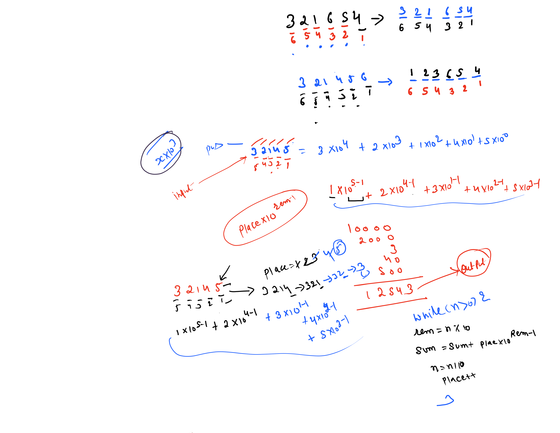
Take the following as input.

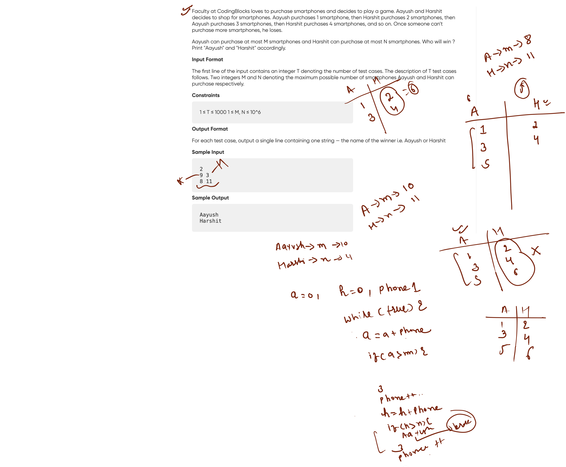
A number

Assume that for a number of n digits, the value of each digit is from 1 to n and is unique. E.g. 32145 is a valid input number.

Write a function that returns its inverse, where inverse is defined as follows

Inverse of 32145 is 12543. In 32145, "5" is at 1st place, therefore in 12543, "1" is at 5th place; in 32145, "4" is at 2nd place, therefore in 12543, "2" is at 4th place.





Faculty at CodingBlocks loves to purchase smartphones and decides to play a game. Aayush and Harshit decides to shop for smartphones. Aayush purchases 1 smartphone, then Harshit purchases 2 smartphones, then Aayush purchases 3 smartphones, then Harshit purchases 4 smartphones, and so on. Once someone can't purchase more smartphones, he loses.

Aayush can purchase at most M smartphones and Harshit can purchase at most N smartphones. Who will win ? Print "Aayush" and "Harshit" accordingly.

Input Format

The first line of the input contains an integer T denoting the number of test cases. The description of T test cases follows. Two integers M and N denoting the maximum possible number of smartphones Aayush and Harshit can purchase respectively.

Constraints

```
1 ≤ T ≤ 1000 1 ≤ M, N ≤ 10^6
```

Output Format

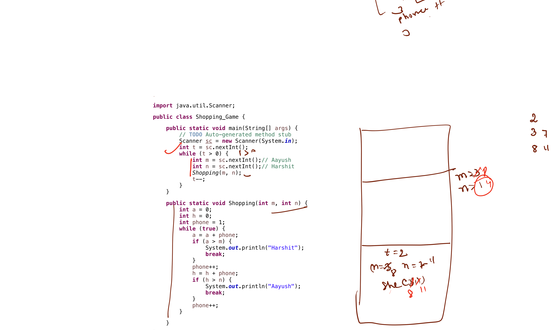
For each test case, output a single line containing one string -- the name of the winner i.e. Aayush or Harshit

Sample Input

```
2
3 3
8 11
```

Sample Output

```
Aayush
Harshit
```



```java
import java.util.Scanner;
public class Shopping_Game {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        int t = sc.nextInt();
        while (t > 0) {
            int m = sc.nextInt();// Aayush
            int n = sc.nextInt();// Harshit
            Shopping(m, n);
            t--;
        }
    }

    public static void Shopping(int n, int n) {
        int a = 0;
        int h = 0;
        int phone = 1;
        while (true) {
            a = a + phone;
            if (a > n) {
                System.out.println("Harshit");
                break;
            }
            phone++;
            h = h + phone;
            if (h > n) {
                System.out.println("Aayush");
                break;
            }
            phone++;
        }
    }
}
```